

Differential Equations in Geophysical Fluid Dynamics

XXVII. Coastal upwelling using reduced gravity model

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Recap

Coastal upwelling based on the reduced gravity model is given by

Governing equation:

$$-f\bar{v}^{(1)} = -g' \frac{\partial H_1}{\partial x} \quad (1a)$$

$$\frac{\partial \bar{v}^{(1)}}{\partial t} + f\bar{u}^{(1)} = \frac{\tau_y^s(t)}{\rho_0 h_1} \quad (1b)$$

$$\frac{\partial H_1}{\partial t} + h_1 \frac{\partial \bar{u}^{(1)}}{\partial x} = 0 \quad (1c)$$

Boundary conditions:

$$\bar{u}|_{x=0} = \bar{v}|_{x=0} = 0 \quad (1d)$$

$$\lim_{x \rightarrow \infty} H_1 = h_1 \quad (1e)$$

Geostrophic adjustment

$$\boxed{\frac{g' h_1}{f^2} \frac{\partial^2 H_1}{\partial x^2} - H_1 = -h_1} \quad (2a)$$

Wind impulse

$$\left. \frac{g'}{f} \frac{\partial H_1}{\partial x} \right|_{x=0} = \int_t^0 \frac{\tau_y^s}{\rho_0 h_1} dt \quad (2b)$$

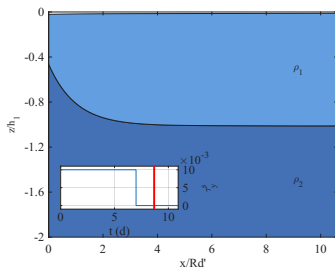
$$\lim_{x \rightarrow \infty} H_1 = h_1 \quad (2c)$$

Recap

Solution to the problem is given by

$$H = h_1 - \left(\frac{1}{Rd'} \int_0^t \frac{\tau_y^s}{\rho_0 f} dt \right) e^{-x/Rd'} \quad (3)$$

where Rd' is the internal Rossby radius of deformation and corresponds to the cross-shore length scale of the coastal upwelling.



https://jang-geun.github.io/vis_upwll1.gif

Limitation

There are many misnomers about the theoretical model discussed by Cushman-Roisin and Beckers (2011) who stated:

*“The lower layer is, by virtue of the choice of a reduced-gravity model, **infinitely deep and motionless**. In the absence of longshore variations, ...”*

Even if those assumptions are alleviated (e.g., Yoshida, 1955), it still assumes **stratified and frictionless** ocean with **flat bottom**¹.

$$\left. \frac{g'}{f} \frac{\partial H_1}{\partial x} \right|_{x=0} = \int_t^0 \frac{\tau_y^s}{\rho_0 h_1} dt \quad (4)$$

$$H = h_1 - \left(\frac{1}{Rd'} \int_0^t \frac{\tau_y^s}{\rho_0 f} dt \right) e^{-x/Rd'} \quad (5)$$

¹What if there is constant wind stress? What does wind impulse physically represent?

Limitation

For a shallow frictional coastal ocean over a slope, traditional theories based on layered models (Yoshida, 1955; Csanady, 1981; Cushman-Roisin, O'Brien, and Smith, 1983; Cushman-Roisin and Beckers, 2011) do not work, but typical (single layer) shallow water equation can be used (Pringle, 2002; Estrade et al., 2008).

$$-f\bar{v} = -g\frac{\partial\eta}{\partial x} \quad (6a)$$

$$f\bar{u} = \frac{\tau_y^s}{\rho_0 h} - \frac{\tau_y^b}{\rho_0 h} \quad (6b)$$

$$\frac{\partial\eta}{\partial t} + \frac{\partial(h\bar{u})}{\partial x} = 0 \quad (6c)$$

$$\bar{u}|_{x=0} = 0 \quad (\text{boundary condition}) \quad (6d)$$

- ▶ Theories unifying both limits: Lentz and Chapman, 2004; Choi, Pringle, and Lippmann, 2023
- ▶ Limitation of two-dimensional (cross-sectional) models: Li and Ruan, 2024; He and Mahadevan, 2021

Two layer ocean model

$$\frac{\partial \bar{u}^{(1)}}{\partial t} - f \bar{v}^{(1)} = -g \frac{\partial \eta}{\partial x} \quad (7a)$$

$$\frac{\partial \bar{v}^{(1)}}{\partial t} + f \bar{u}^{(1)} = -g \frac{\partial \eta}{\partial y} \quad (7b)$$

$$\frac{\partial \eta}{\partial t} + h_1 \left(\frac{\partial \bar{u}^{(1)}}{\partial x} + \frac{\partial \bar{v}^{(1)}}{\partial y} \right) - \frac{\partial \eta_2}{\partial t} = 0 \quad (7c)$$

$$\frac{\partial \bar{u}^{(2)}}{\partial t} - f \bar{v}^{(2)} = -g \frac{\partial \eta}{\partial x} - g' \frac{\partial \eta_2}{\partial x} \quad (8a)$$

$$\frac{\partial \bar{v}^{(2)}}{\partial t} + f \bar{u}^{(2)} = -g \frac{\partial \eta}{\partial y} - g' \frac{\partial \eta_2}{\partial y} \quad (8b)$$

$$\frac{\partial \eta_2}{\partial t} + h_2 \left(\frac{\partial \bar{u}^{(2)}}{\partial x} + \frac{\partial \bar{v}^{(2)}}{\partial y} \right) = 0 \quad (8c)$$

The system of differential equations for a two-layer ocean model can be handled in the same manner, but it is too complicated to provide good insight.

The use of some useful approximations (e.g., $\Delta\rho/\rho_0 \approx 0$ and $\partial\eta/\partial t \approx 0$) is tricky.

Decoupling systems by defining barotropic and baroclinic components

To decouple the system, we can define new velocity variables:

$$\bar{u} = (h_1 \bar{u}^{(1)} + h_2 \bar{u}^{(2)})/h, \quad (9a) \quad \bar{v} = (h_1 \bar{v}^{(1)} + h_2 \bar{v}^{(2)})/h, \quad (9b)$$

$$\bar{u}' = \bar{u}^{(2)} - \bar{u}^{(1)} \quad (10a) \quad \bar{v}' = \bar{v}^{(2)} - \bar{v}^{(1)} \quad (10b)$$

Vertical averaging of the equations² yields equations for barotropic mode ($\bar{u}$, $\bar{v}$, and η).

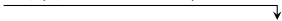
Subtracting the surface layer equation from the subsurface equations³ yields equations for baroclinic mode ($\bar{u}'$, $\bar{v}'$, and η_2).

² $(h_1 \times (7a) + h_2 \times (8a))/h$, $(h_1 \times (7b) + h_2 \times (8b))/h$, and $((7c) + (8c))/h$

³ $(8a) - (7a)$, $(8b) - (7b)$, and $(8c)/h_2 - (7c)/h_1$

Decoupling systems by defining barotropic and baroclinic components

$$\Delta\rho/\rho_0 \ll 1 \text{ or } h_2/h \ll 1$$



$$-f\bar{v} = -g\frac{\partial\eta}{\partial x} - g'\frac{h_2}{h}\frac{\partial\eta_2}{\partial x} \quad (11a)$$

$$\frac{\partial\bar{v}}{\partial t} + f\bar{u} = 0 \quad (11b)$$

$$\frac{\partial\eta}{\partial t} + h\frac{\partial\bar{u}}{\partial x} = 0 \quad (11c)$$

$$-f\bar{v}' = -g'\frac{\partial\eta_2}{\partial x} \quad (12a)$$

$$\frac{\partial\bar{v}'}{\partial t} + f\bar{u}' = 0 \quad (12b)$$

$$-\frac{h_2}{h}\frac{\partial\eta}{\partial t} + \frac{\partial\eta_2}{\partial t} + h'\frac{\partial\bar{u}'}{\partial x} = 0$$

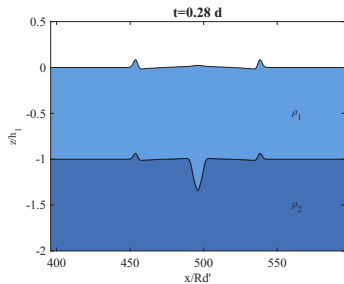
$\uparrow$
 $h_2/h \ll 1 \text{ or } L \ll Rd$

(12c)

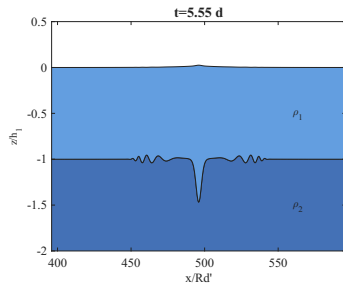
where $h' = h_1 h_2 / h$ is referred to as equivalent depth.

Now we have decoupled systems, (11) and (12)!

Barotropic and baroclinic modes



(a)



(b)

https://jang-geun.github.io/vis_two_layer_wave.gif

References I

-  Choi, Jang-Geun, James Pringle, and Thomas Lippmann (2023). “A perturbative solution for nonlinear stratified upwelling over a frictional slope”. In: *Journal of Physical Oceanography* 53.10, pp. 2317–2330.
-  Csanady, G. T. (1981). “Circulation in the coastal ocean”. In: *Advances in geophysics*. Vol. 23. Elsevier, pp. 101–183.
-  Cushman-Roisin, Benoit and Jean-Marie Beckers (2011). *Introduction to geophysical fluid dynamics: physical and numerical aspects*. Vol. 101. Academic press.
-  Cushman-Roisin, Benoit, James J O'Brien, and Robert L Smith (1983). “On wind and ocean-velocity correlations in a coastal-upwelling system”. In: *Journal of Physical Oceanography* 13.3, pp. 547–550.

References II

-  Estrade, Philippe et al. (2008). “Cross-shelf structure of coastal upwelling: a two-dimensional extension of Ekman’s theory and a mechanism for inner shelf upwelling shut down”. In: *Journal of marine research* 66.5, pp. 589–616.
-  He, Jing and Amala Mahadevan (2021). “How the source depth of coastal upwelling relates to stratification and wind”. In: *Journal of Geophysical Research: Oceans* 126.12, e2021JC017621.
-  Lentz, Steven J and David C Chapman (2004). “The importance of nonlinear cross-shelf momentum flux during wind-driven coastal upwelling”. In: *Journal of Physical Oceanography* 34.11, pp. 2444–2457.
-  Li, Dou and Xiaozhou Ruan (2024). “On the pathways of wind-driven coastal upwelling: Nonlinear momentum flux and baroclinic instability”. In: *Journal of Physical Oceanography* 54.1, pp. 63–79.

References III

-  Pringle, James M (2002). "Enhancement of wind-driven upwelling and downwelling by alongshore bathymetric variability". In: *Journal of Physical Oceanography* 32.11, pp. 3101–3112.
-  Yoshida, Kozo (1955). "Coastal upwelling off the California coast.". In: *Rec. Oceanogr. Wks. Japan, New Ser. 2*, pp. 8–20.